

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
In [8]: import os
print("current folder:", os.getcwd())
print("files in this folder:", os.listdir())
```

current folder: c:\Users\osfra\OneDrive\Desktop\marketing analysis
files in this folder: ['.git', 'marketing_analysis.ipynb']

```
In [11]: df = pd.read_csv(r"c:\Users\osfra\OneDrive\Desktop\marketing_ROI_Analysis.csv")
df.head()
```

```
Out[11]:
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

```
In [12]: df.shape
```

```
Out[12]: (4572, 4)
```

```
In [13]: df.columns
```

```
Out[13]: Index(['TV', 'Radio', 'Social_Media', 'Sales'], dtype='str')
```

```
In [14]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   TV              4562 non-null   float64
1   Radio           4568 non-null   float64
2   Social_Media    4566 non-null   float64
3   Sales           4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

```
In [15]: df.describe()
```

Out[15]:

	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

In [16]: `df.isnull().sum()`

Out[16]: TV 10
 Radio 4
 Social_Media 6
 Sales 6
 dtype: int64

In [17]: `df = df.dropna()`In [18]: `df.isnull().sum()`

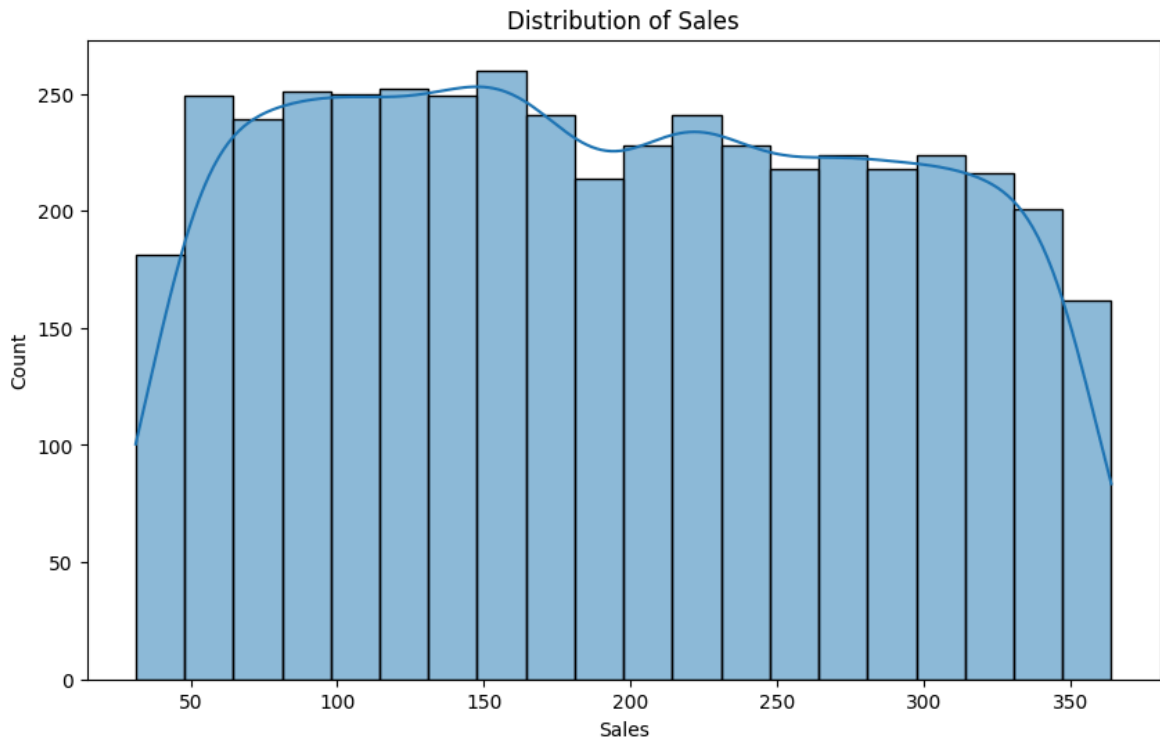
Out[18]: TV 0
 Radio 0
 Social_Media 0
 Sales 0
 dtype: int64

EPLORARY DATA ANALYSIS EDA \

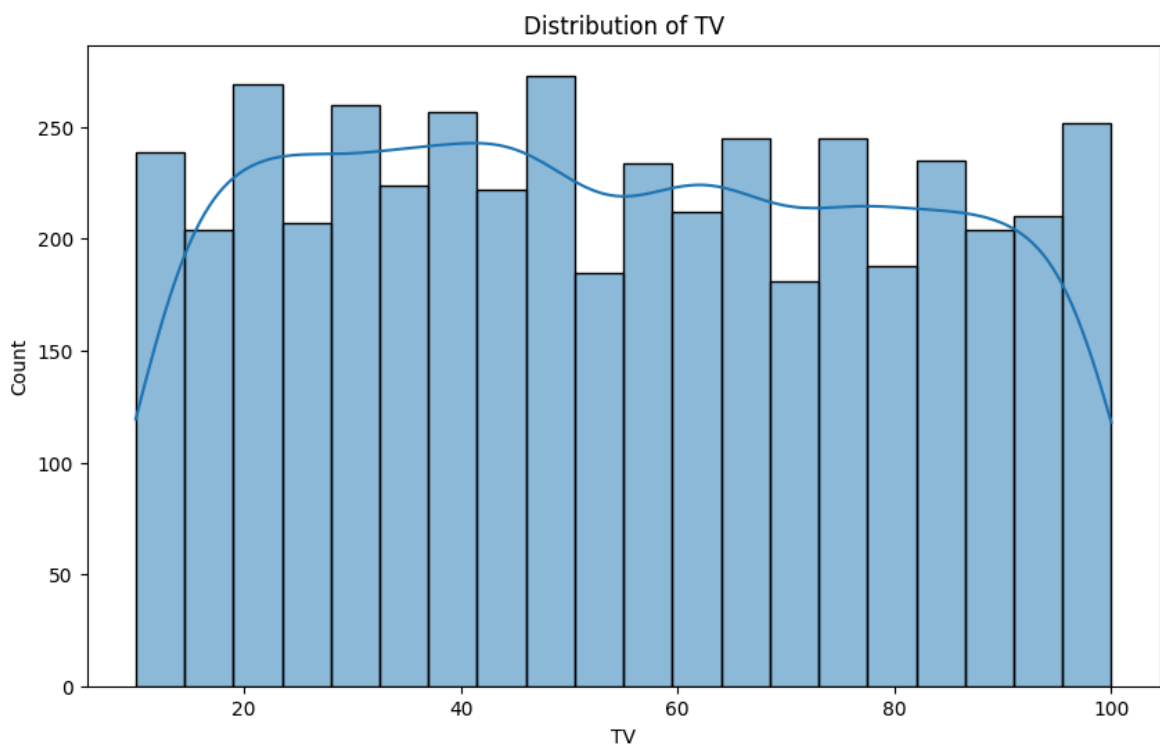
DISTRIBUTION OF SALES

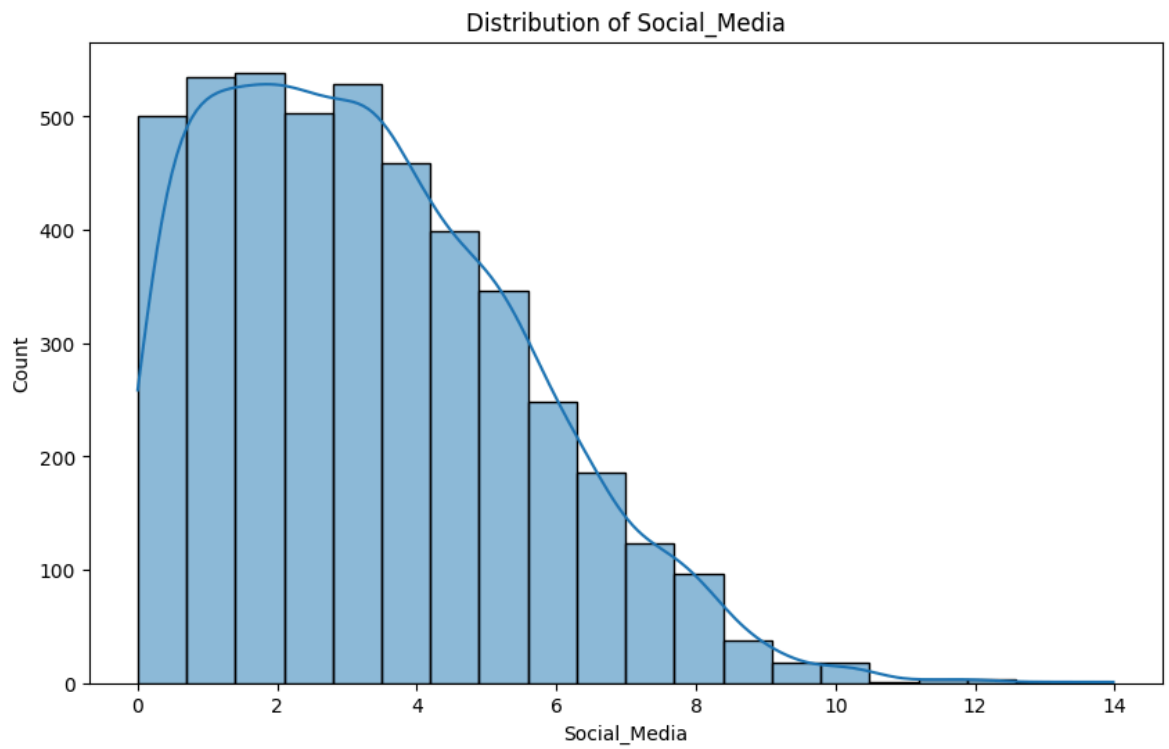
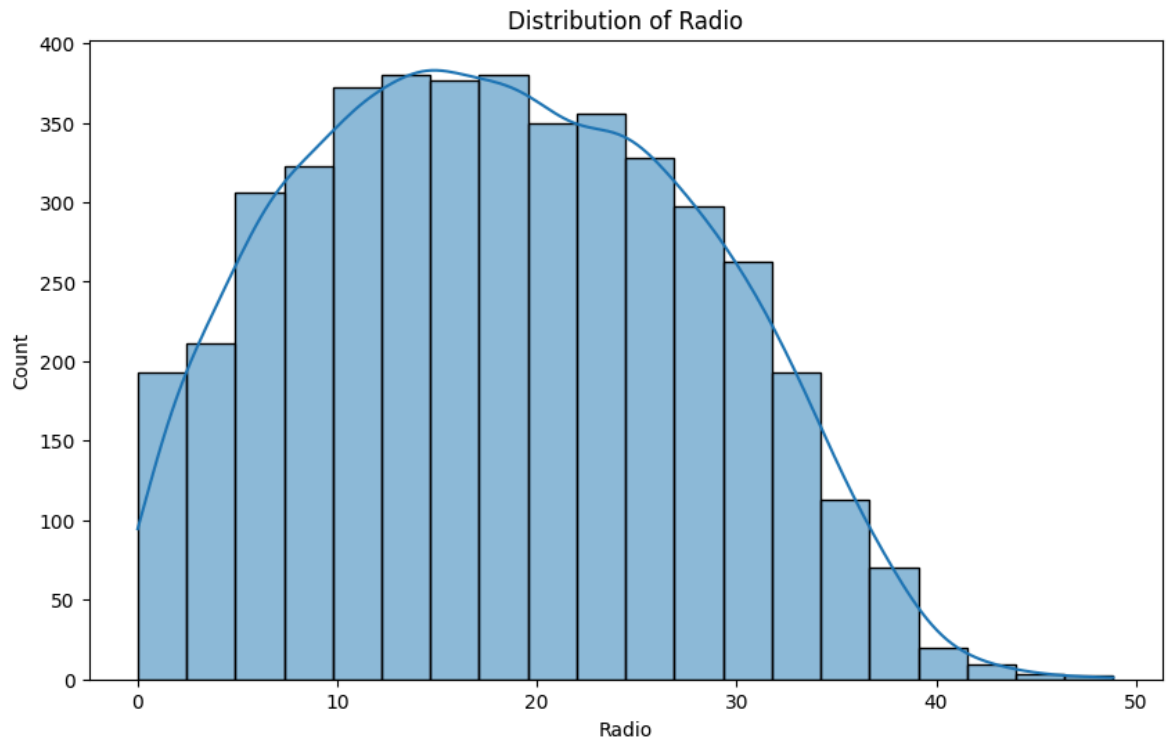
In [21]: `plt.figure(figsize=(10, 6))
 sns.histplot(df['Sales'], bins=20, kde=True)
 plt.title('Distribution of Sales')`

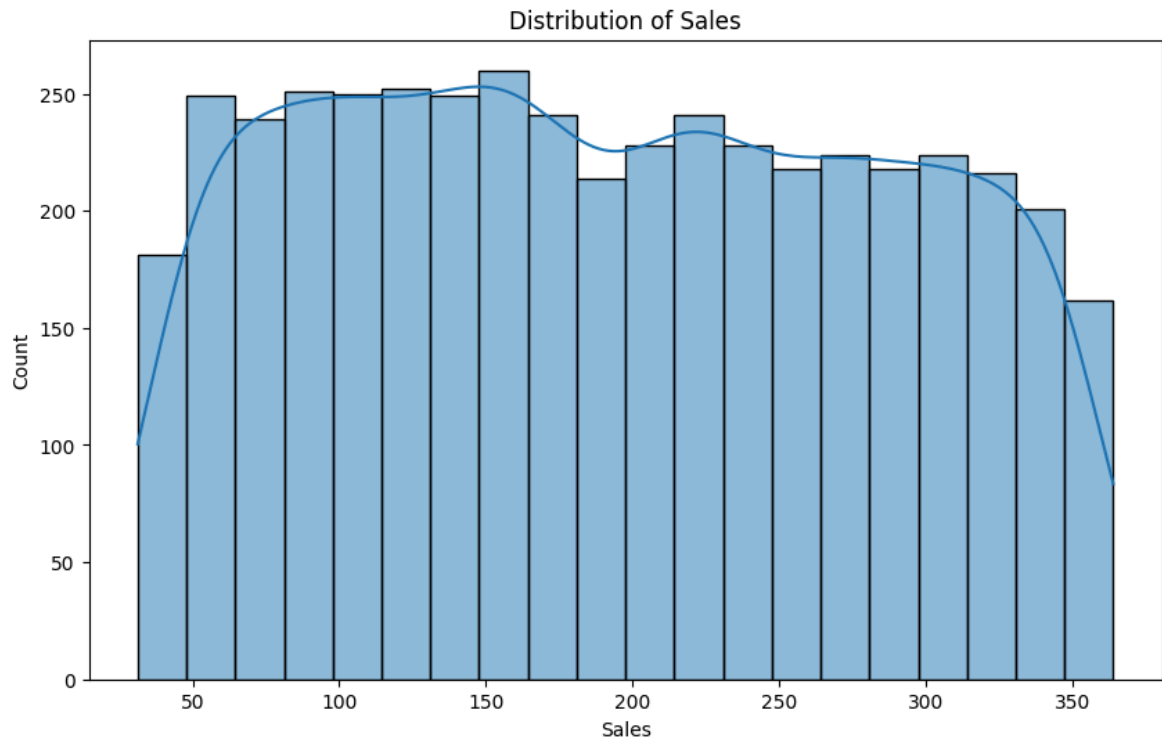
Out[21]: Text(0.5, 1.0, 'Distribution of Sales')



```
In [22]: for col in df.select_dtypes(include=['int64', 'float64']).columns:
plt.figure(figsize=(10, 6))
sns.histplot(df[col], bins=20, kde=True)
plt.title(f'Distribution of {col}')
plt.show()
```

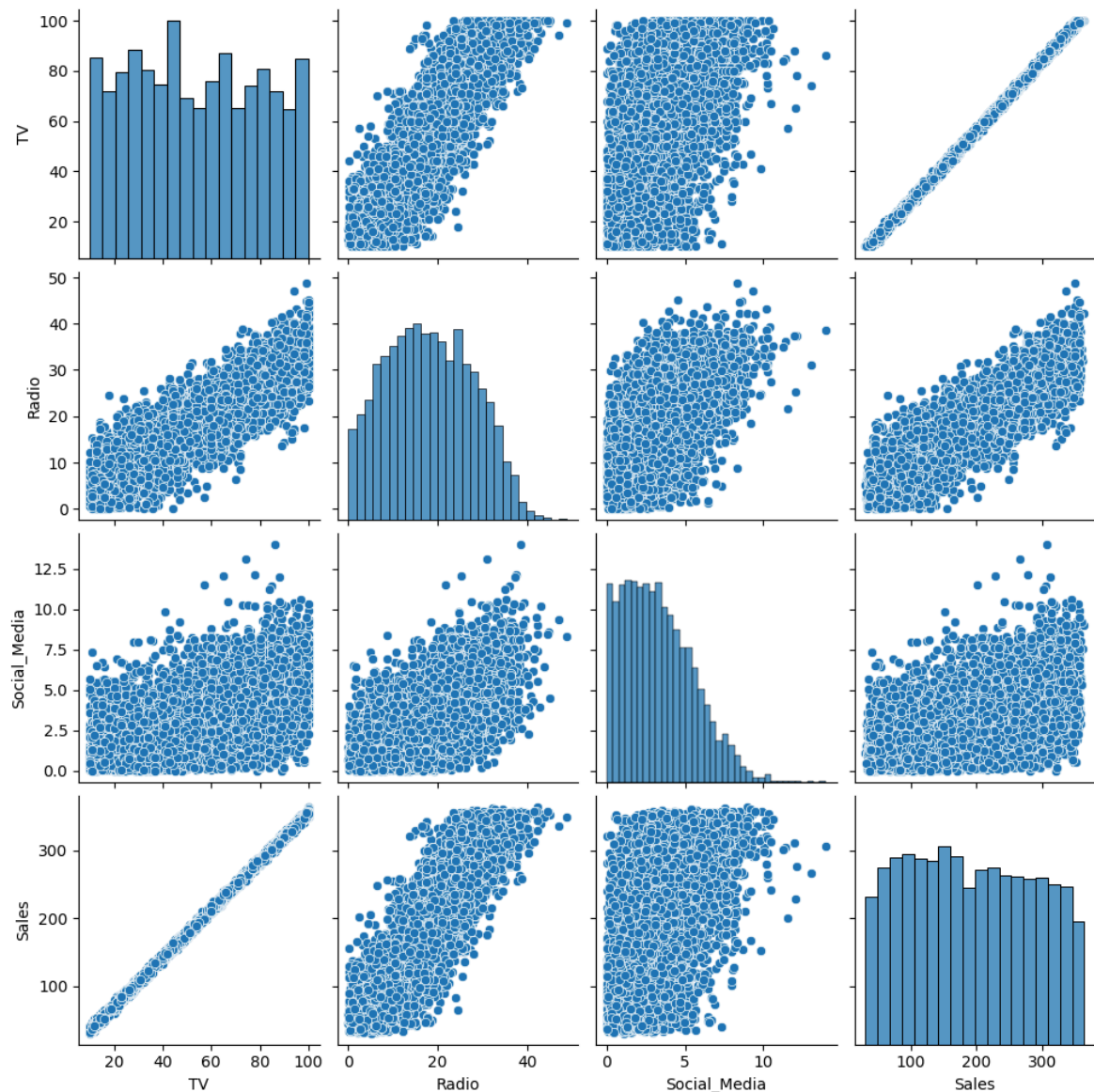






Relationship between variables

```
In [23]: sns.pairplot(df)
plt.show()
```



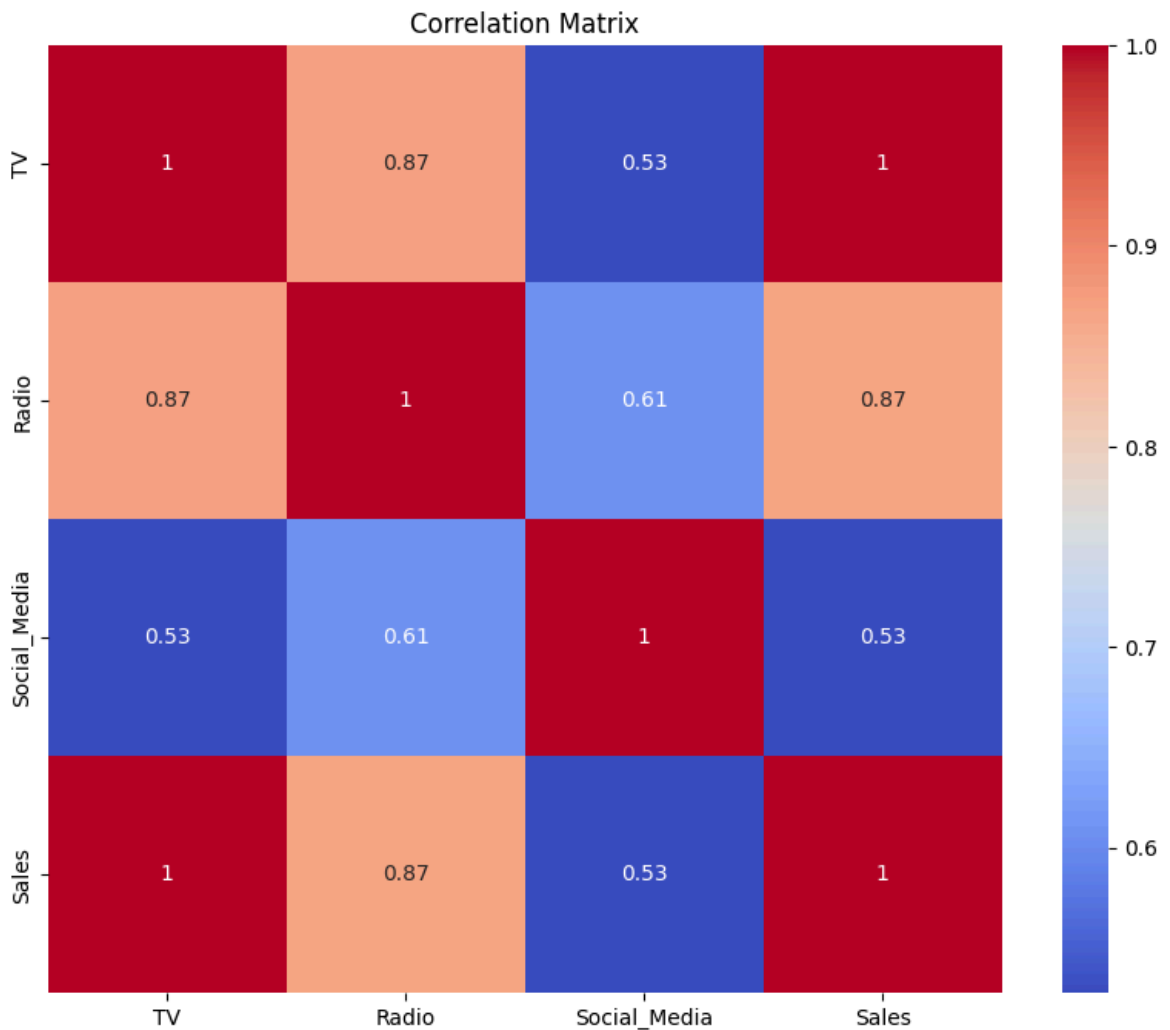
CORRELATION ANALYSIS

```
In [24]: corr = df.corr(numeric_only=True)
```

SHOW HEATMAP

```
In [25]: plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
```

```
Out[25]: Text(0.5, 1.0, 'Correlation Matrix')
```



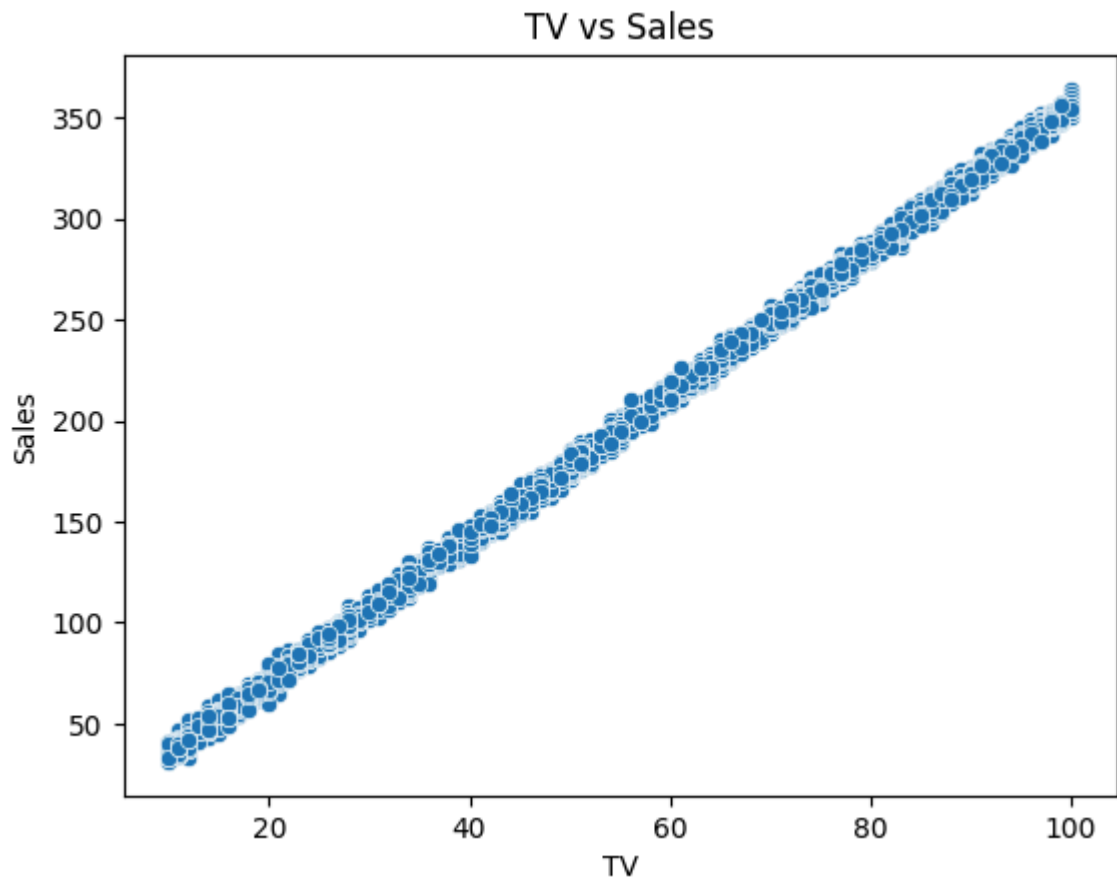
check correlation with sales

```
In [28]: corr['Sales'].sort_values(ascending=False)
```

```
Out[28]: Sales      1.000000
TV          0.999497
Radio       0.868638
Social_Media 0.527446
Name: Sales, dtype: float64
```

SINCE TV HAS THE HIGHEST SALES SO IT IS THE INDEPENDENT VARIABLE

```
In [29]: sns.scatterplot(x='TV', y='Sales', data=df)
plt.title('TV vs Sales')
plt.show()
```



Building a regression model by defining X and Y

```
In [30]: x = df[['TV']]  
         y = df[['Sales']]
```

ADD A CONSTANT

```
In [31]: x = sm.add_constant(x)
```

FIT MODEL

```
In [33]: model = sm.OLS(y, x).fit()  
         print(model.summary())
```


OLS Regression Results

=====						
Dep. Variable:	Sales		R-squared:	0.999		
Model:	OLS		Adj. R-squared:	0.999		
Method:	Least Squares		F-statistic:	4.517e+06		
Date:	Sun, 07 Jun 2026		Prob (F-statistic):	0.00		
Time:	17:53:44		Log-Likelihood:	-11366.		
No. Observations:	4546		AIC:	2.274e+04		
Df Residuals:	4544		BIC:	2.275e+04		
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565
=====						
Omnibus:	0.052		Durbin-Watson:	1.998		
Prob(Omnibus):	0.974		Jarque-Bera (JB):	0.031		
Skew:	-0.001		Prob(JB):	0.985		
Kurtosis:	3.012		Cond. No.	138.		
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

From the OLS regression performed to examine how TV advertising budget affects sales,

the dependent variable = sales independent variable = Tv adveetising spending number of observations 4,546 key finding

1. MODEL PERFORMANCE

R- squared = 0.999: This means 99.9% of the variation in sales is explained by Tv advert spending which is almost a perfect fit model

2. TV Advertising Coeficient

coefficient = 3.5615 (p-value <0.001) for every
1 increase in TV advertising budget, sales increased by approximately 3.56 on average. This relationship is highly highly significant and statistical

3. INTERCEPT

0.1325(not statistically significant) when TV advert spend is zero, predicted sales are close to zero

4. OVERALL MODEL SIGNIFICANCE

F- statistic is extremely high with p-value 0.000 the model as a whole is highly significant.

IN SUMMARY

The OLS regression result as a whole, demonstrate that Tv advertising is powerful driver sales. With an R- squared of 0.999, the model shows that TV spend explains nearly all variations in sales. The coefficient of 3.5616 indicates a strong positive relationship suggesting that increasing TV budget is likely to generate substantial returns. This finding supports allocating more marketing resources towards TV advertising.

PREDICTED VALUES

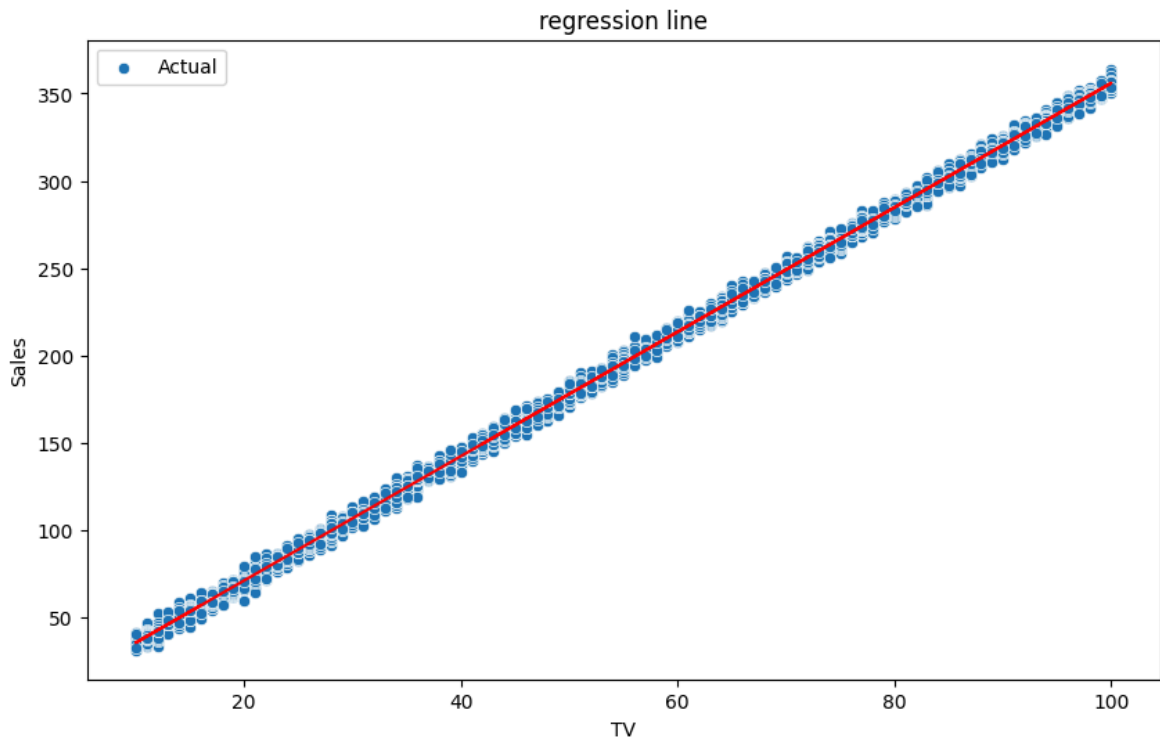
```
In [37]: df['predicted_Sales'] = model.predict(x)
df[['Sales', 'predicted_Sales']].head()
```

```
Out[37]:
```

	Sales	predicted_Sales
0	54.732757	56.851733
1	46.677897	46.167191
2	150.177829	145.889585
3	298.246340	295.473177
4	56.594181	53.290219

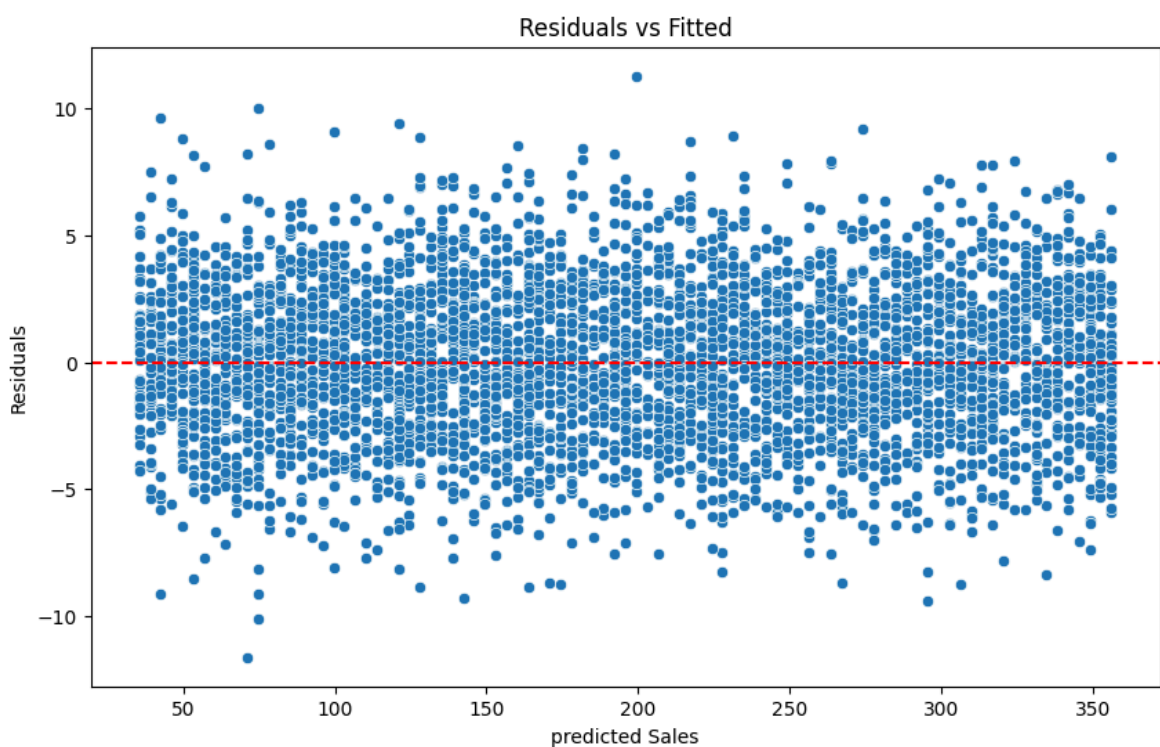
REGRESSION LINE PLOT

```
In [39]: plt.figure(figsize=(10,6))
sns.scatterplot(x=df['TV'], y=df['Sales'], data=df, label='Actual')
plt.plot(df['TV'], df['predicted_Sales'], color='red', label='Predicted')
plt.title("regression line")
plt.show()
```



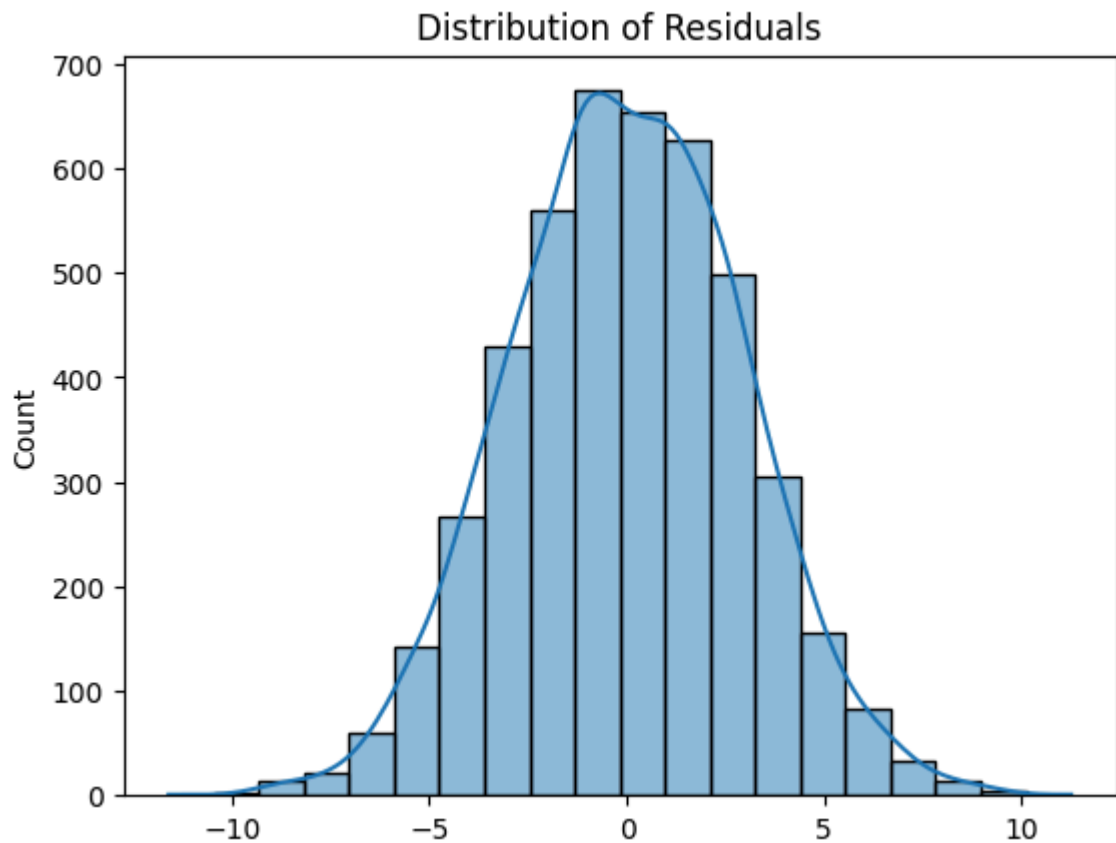
DIAGNOSTIC PLOT LINEARITY

```
In [42]: residuals = model.resid
plt.figure(figsize=(10,6))
sns.scatterplot(x=df['predicted_Sales'], y=residuals)
plt.axhline(0, color='red', linestyle='--')
plt.title('Residuals vs Fitted')
plt.xlabel('predicted Sales')
plt.ylabel('Residuals')
plt.show()
```



DIAGNOSTIC PLOT NORMALITY

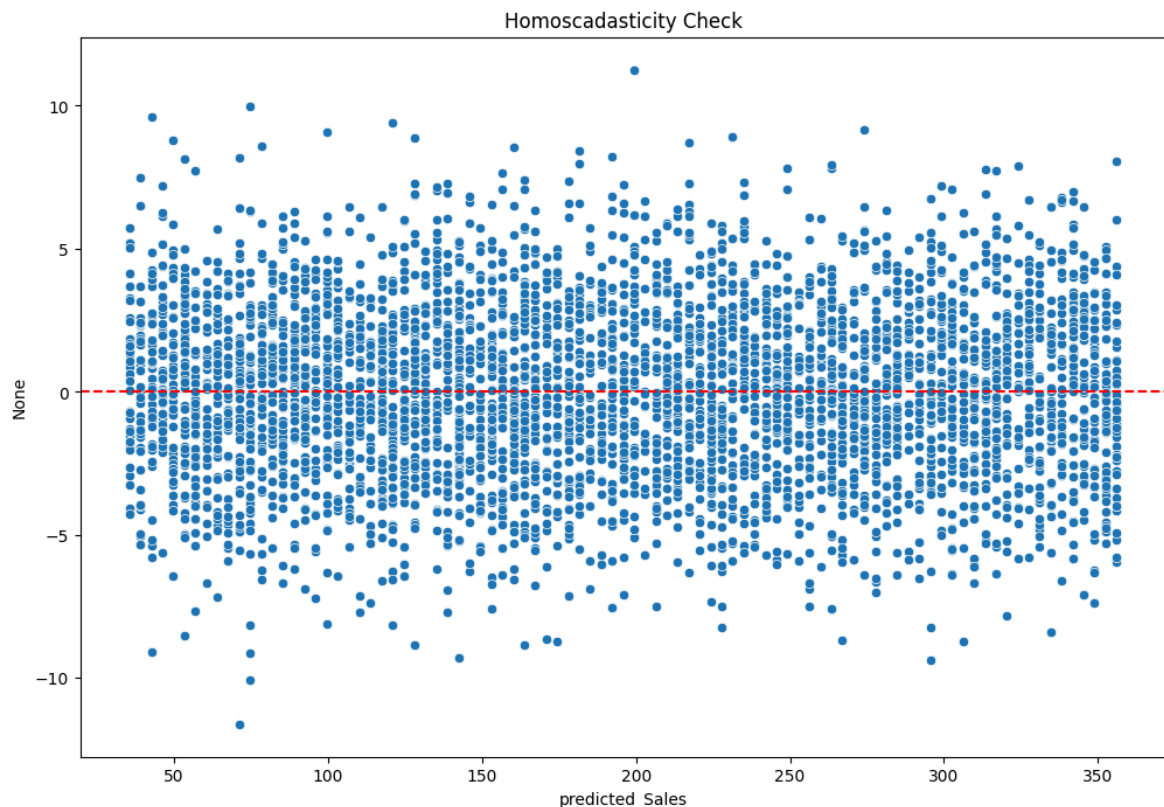
```
In [43]: sns.histplot(residuals, bins=20, kde=True)
plt.title('Distribution of Residuals')
plt.show()
```



Residuals approximately follows a normal distribution

DIAGNOSTIC PLOT- HOMOSCEDASTICITY

```
In [44]: plt.figure(figsize=(12, 8))
sns.scatterplot(x=df['predicted_Sales'], y=residuals)
plt.axhline(0, color='red', linestyle='--')
plt.title("Homoscedasticity Check")
plt.show()
```



BUSINESS INTERPRETATION

The regression analysis shows that Tv advertising has the strongest relationship with sales. The model achieved an R-squared of 0.99, indicating that 99% of sales variation is explained by TV advert expenditure. The coefficient for TV advert is positive and statistically significant, suggesting that increased TV advert spending leads to higher sales. Diagnostic plots indicate that the assumptions of linear regression are reasonably satisfied.

ROI RECOMMENDATION

Based on the correlation analysis and regression results, TV advertising should receive the highest marketing budget allocation. TV has the strongest positive relationship with sales and provides the most reliable prediction of revenue growth. Therefore, management should prioritize investment in TV campaigns to maximize marketing ROI.